Ex-No: 6

12.8.26

Hamming Code

Aim:

Write a program to implement error detection and correction using hamming code concept. Make test run to input data stream and verify correction feature.

CODE:

```
def calc-redundant-bits(m):
    for r in range(m+1):
        if(2**r) >= (m+r+1):
            return r

def main():
    data = input().strip()
    m = un(data)
    r = calc-redundant-bits(m)
    n = m + r
    print(f"Number of r needed: {r}")
    print(f"Total frame length: {n}")
    code = ['0'] *(n+1)
    j=0
    for i in range(1,n+1):
        if(i &(i-1)) !=0:
            code[i] = data [-(j+1)]
            j+=1
    for i in range(r):
        parity-pos = 2** i
```

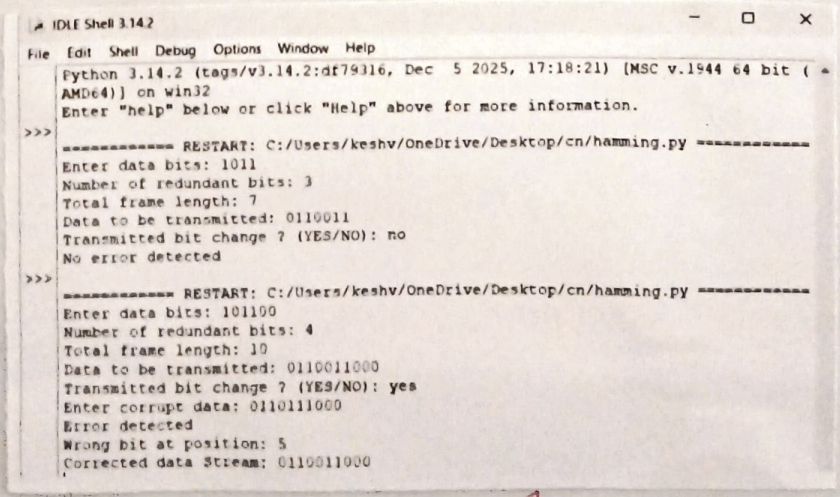
```
val=0
for pos in range(1,n+1):
    if(pos & parity_pos)!=0 and
            pos!=parity_pos:
        if code[pos]=='1':
            val^=1
Code[parity_pos]=str(val)
transmitted_str="".join(code[1:]
                        [::-1])
print(f"Data to be transmitted:
        {transmitted_str}")
change=input("\nTransmitted bit change?
            .strip().upper()
if change=="yes":
    received_str=input("Corrupt data?")
                .strip()
    if len(received_str)!=n:
        print(f"Error: received data
            be exactly {n} bits")
        return
    received_code=['0']+list(received_str
                            [::-1])
else:
    received_code=list(code)

error_pos=0
for i in range(r):
    parity_pos=2**i
    val=0
    for pos in range(1,n+1):
        if(pos & parity_pos)!=0:
            if received_code[pos]=='1':
                val^=1
```

```
    if val != 0:
        error_pos += parity_pos

if error_pos == 0:
    print("No error detected")
else:
    print("Error detected")
    print(f"Wrong bit at position: {error_pos}")
    received_code[error_pos] = '0' if received_code[error_pos] == '1' else '1'
    corrected_str = "".join(received_code[1:][::-1])
    print(f"Corrected data Stream: {corrected_str}")

if __name__ == "__main__":
    main()
```



output:

① Enter data bits: 101100
Number of redundant bits : 4
Total frame length : 10
Data to be transmitted : 1011100011
Transmitted bit change ? (Yes/No) : Yes
Enter Corrupt data : 1011101011
Error detected
Wrong bit at position : 4
Corrected data stream : 1011100011

② Enter data bits = 1011
Number of redundant bits = 3
Total data frame length : 7
Data to be Transmitted : ~~1010101~~ 0110011
Transmitted bit Change ? (Yes/No) : No
No error detected.

Result:
Thus, the hamming code was implemented successfully for error detection and correction.